

|          |                                                                                                                                         |                |     |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------|----------------|-----|
| <b>9</b> | <b>(a)</b> nucleus/nuclei emits<br>spontaneously/randomly<br>$\alpha$ -particles, $\beta$ -particles, $\gamma$ -ray photons             | M1<br>A1<br>A1 | [3] |
|          | <b>(b) (i)</b> $N - \Delta N$                                                                                                           | A1             | [1] |
|          | <b>(ii)</b> $\Delta N / \Delta t$                                                                                                       | A1             | [1] |
|          | <b>(iii)</b> $\Delta N / N$                                                                                                             | A1             | [1] |
|          | <b>(iv)</b> $\Delta N / N \Delta t$                                                                                                     | A1             | [1] |
|          | <b>(c)</b> graph: smooth curve in correct direction starting at (0,0)<br>$n$ at $2t_{1/2}$ is 1.5 times that at $t_{1/2}$ ( $\pm 2$ mm) | M1<br>A1       | [2] |

|               |                                                           |                 |              |
|---------------|-----------------------------------------------------------|-----------------|--------------|
| <b>Page 5</b> | <b>Mark Scheme</b>                                        | <b>Syllabus</b> | <b>Paper</b> |
|               | <b>Cambridge International AS/A Level – May/June 2015</b> | <b>9702</b>     | <b>41</b>    |

## **Section B**